Sections for ‘Applications of Quantum Computing Across Different Fields’

Finance: Quantum Computing Reduces Risks in Financial Networks

Reference: Aboussalah, A.M., Chi, C. and Lee, C.G., 2023. Quantum computing reduces systemic risk in financial networks. Scientific Reports, 13(1), p.3990.

In finance, quantum computing is used to resolve issues in increasing interdependencies between financial institutions. Inter-dependencies are associated with loans, holding shares, and other liabilities. Institutions are connected by their interdependencies and form a financial network.

Failure in financial institutions can propagate across the industry. Financial institutions need to adjust for these inter-dependencies to prevent non-linear cascade failures which can have detrimental effects on the market. Networks need to optimize connections between institutions. Aboussalah et al. developed new techniques for classical and quantum partitioning for directed and weighted first graphs. Experimental results indicated that quantum partitioning is more robust to financial shocks; delays cascades failure phase transition; reduces total number of failures of convergence under systemic risks with reduced timed complexity. The one-stage optimization algorithm globally optimizes the cross-holdings across the interbank network in order to minimize the total loss. “Using quantum partitioning for our two-stage optimization creates better partitions which further delays the phase transition of cascades and comes with additional computational efficiency benefits due to quantum superposition effects. However, we were only able to test up to 50 organizations due to quantum hardware limitations but quantum hardware will improve over time.”

Applications of Quantum Computing in Biology

References: {Ghamsari, M.S., 2025. Quantum computing applications in biology. Discover Computing, 28(1), p.276.} {Kumar, G., Yadav, S., Mukherjee, A., Hassija, V. and Guizani, M., 2024. Recent advances in quantum computing for drug discovery and development. IEEE Access, 12, pp.64491-64509.} {Lau, B., Emani, P.S., Chapman, J., Yao, L., Lam, T., Merrill, P., Warrell, J., Gerstein, M.B. and Lam, H.Y., 2023. Insights from incorporating quantum computing into drug design workflows. Bioinformatics, 39(1), p.btac789.}

Quantum biology is the area of study that seeks to elucidate biological processes that classical physics and computational techniques cannot fully explain, providing evidence for the presence of quantum phenomena in living organisms. Examples include avian navigation using the Earth's magnetic field, photosynthesis, and genomic processes such as DNA transmutation. Quantum effects such as coherence, tunneling and entanglement. Coherence is believed to allow coordination and synchronization of individual components within a biological system across various spatial domains.

"Quantum Computing has significant potential to advance sci-ence and technology, particularly in areas like molecular biology. However, in the nearfuture, classical computers will still be essential for many everyday applications. Quan-tum Computing enhances computational drug discovery by improving the accuracyand efficiency of simulations, strengthening machine learning models, and speeding upthe pharmaceutical research process from initial screening, to preclinical evaluation. Italso supports research onto complex diseases, fostering more equitable healthcare andresilience against global health crises."

"Lau et al. discuss the concept of HypaCADD, a hybrid classical-quantum workflow method for determining ligand binding to proteins, and also considers genetic mutations. They discussed how HypaCADD helps combine classical docking and molecular dynamics with Quantum Machine Learning (QML) to get a report on the impact of mutation. This paper outlines a neural network constructed using qubit-rotation gates. It maps a classical machine learning module onto QC. This is explained by taking a case study of the novel coronavirus (SARS-CoV-2) protease and its mutants. This paper also states how QML performs on par with classical computing, if not better. It summarises a successful strategy for leveraging QC for CADD by HypaCADD."

Cybersecurity

References: {Yalcin, H., Daim, T., Moughari, M.M. and Mermoud, A., 2024. Supercomputers and quantum computing on the axis of cyber security. Technology in Society, 77, p.102556.} {Baseri, Y., Chouhan, V. and Ghorbani, A., 2024. Cybersecurity in the quantum era: Assessing the impact of quantum computing on infrastructure. arXiv preprint arXiv:2404.10659.}

“Quantum computing and supercomputers are technologies that have the potential to revolutionize many industries, but also carry significant cybersecurity risks [1,2]. Quantum computers can handle much larger data sets than even the world's most powerful supercomputers based on classical computing, while they have the power to perform much more complex calculations [3]. Although quantum computers are still in their infancy, they have the potential to be considered in terms of cybersecurity threats.”

“Organizations are actively researching and developing quantum-resistant encryption methods and security protocols to mitigate these potential risks. Shor’s algorithm directly breaks RSA and the Digital Signature Standard in polynomial time. Variations of the same idea break elliptic curve cryptography and more general class-group systems in polynomial time.

For comparison, RSA and elliptic curve cryptography are believed to be secure when the attacker is not equipped with a quantum computer. ”

Quantum Computing in Chemistry and Chemical Engineering

References: {Weidman, J.D., Sajjan, M., Mikolas, C., Stewart, Z.J., Pollanen, J., Kais, S. and Wilson, A.K., 2024. Quantum computing and chemistry. Cell Reports Physical Science, 5(9).} {Imoto, T., Susa, Y., Miyazaki, R. and Matsuzaki, Y., 2024. Universal quantum computation using quantum annealing with the transverse-field Ising Hamiltonian. arXiv preprint arXiv:2402.19114.}

“Though classical computers have had a monumental impact on science, there are still many types of computational tasks and problems that remain intractable in terms of scale even for the largest classical machines. In chemistry, for example, the many-body entangled states underlying the electronic structure of a molecule cannot be modeled fully even on today’s most powerful computers. Instead, numerous approximations are required to make computations feasible for larger molecules. These approximations can miss important electronic or electron-nuclear effects, especially for complicated molecular systems such as complexes involving transition metals or heavy elements.”

Imoto et al. present a practice method for implementing universal quantum computation within the conventional quantum annealing architecture using the transverse-field Ising Hamiltonian. Method relies on an adiabatic transformation of the Hamiltonian, changing from transverse fields to a ferromagnetic interaction regime where the ground states become degenerate. “we remark that arbitrary unitary quantum gates can be constructed by combining several types of unitary gates, called universal gate sets. Typically, either phase accumulation (Ramsey interferometry) or application of an oscillating field (such as Rabi oscillations) is adopted to implement the gate operations for a solid-state quantum computer. It has the potential for a wide array of applications, including quantum chemistry, machine learning, financial engineering, optimization problems, and decrypting ciphers. Most quantum algorithms to achieve quantum speed-up need quantum error collection, exemplified by a surface code. This paradigm, known as fault-tolerant quantum computation(FTQC), requires thousands of physical qubits per logical qubit to effectively suppress the noise effect. Consequently, it is estimated that FTQC will at least require 10 million qubits to perform computations beyond the capabilities of existing classical computers. Unlike the previous approach to perform the gate operations by either applying an oscillating field or accumulating a relative phase, we utilize the degenerate ground states to perform the X-rotation gate and controlled-not gate. We demonstrate the effectiveness of our method by numerical simulations and experimental results. Importantly, we do not use microwave pulses to implement gate operations. This feature enhances scalability, as evidenced by the large-scale devices developed by D-Wave Inc. Employing only the transverse-field Ising model, our approach holds substantial promise for developing large-scale gate-based quantum computers.”

Quantum Computing in Physics and Electrical Engineering

References: {Ullah, M.H., Eskandarpour, R., Zheng, H. and Khodaei, A., 2022. Quantum computing for smart grid applications. IET Generation, Transmission & Distribution, 16(21), pp.4239-4257.} {Kumar, M.S., Krishna, R.V.V., Satyanarayana, V. and Ranjit, P.S., 2024. Battery Revolution: Quantum Computing and Machine Intelligence Synergy. In Real-World Challenges in Quantum Electronics and Machine Computing (pp. 30-42). IGI Global Scientific Publishing.} {Khalid, A., Tufail, S. and Sarwat, A.I., 2021. A review on quantum computing approach for Next-Generation energy storage solution. SoutheastCon 2021, pp.1-7.}

“The electric utility companies are slowly advancing towards

employing QC in various applications. Enel, an Italian multinational electricity and gas distributor, made a partnership with

Data Reply, a company that offers AI-powered advanced data

analytics, to solve combinatorial optimization problems based on the quadratic unconstrained binary optimization (QUBO)

model that creates an optimal plan for assigning maintenance

work for the crews. A UK-based utility company E.ON partnered with IBM, and

aims to implement quantum solutions for decentralized energy

infrastructure with DERs… A power flow study, also called load-flow analysis, is a steadystate analysis to compute the currents, voltages, and real,

reactive, and apparent power flows in a system under specified

load settings. Power flow analyses are essential for the efficient

management of electric power systems. Describes a

fast decoupled power flow algorithm based on a quantum state

that utilizes Hermitian and constant Jacobian matrices. Also,

a computationally improved Harrow-Hassidim-Lloyd (HHL)

algorithm is presented to deal with the fast decoupled power

flow by quantum phase estimation and reciprocal rotation. A DC power flow problem is solved using the HHL

algorithm, and it was demonstrated with a faster convergence

rate that the grid security can be enhanced using QC techniques.

However, the next generation quantum computers with large

quantum depths, qubit numbers, long coherence time, and lower

noises may be required to apply such approaches in large-scale

power systems, as suggested in… The current research in QC for smart grid

applications can be classified into two main categories. Category

1 concerns implementing classical algorithms in conventional

computers replicating quantum mechanics fundamentals such

as superposition, entanglement, and error correction and detec-TABLE 3 QC Smart Grid applicationsApplication Category 1 Category 21 Power Flow - [41–44]

2 PMU Placement

3 Unit Commitment

4 Fault Management

5 Smart Grids’ Cyber Security

6 Stability Assessment

7 Transient Analysis

8 Reliability Assessment

9 Facility Location-allocation

10 Power System Control

11 Forecasting

tion. It also includes quantum cryptography, communication,

and simulations. Category 2 involves implementing computationally expensive algorithms on real-time quantum computers.

The literature presented in this paper, summarized in Table 3,

is related to various smart grid applications, which are categorized into the classes mentioned above. However, a detailed

summary of the literature is demonstrated in Table 4.3.1 Power flowA power flow study, also called load-flow analysis, is a steadystate analysis to compute the currents, voltages, and real,

reactive, and apparent power flows in a system under specified

load settings. Power flow analyses are essential for the efficient

management of electric power systems. describes a

fast decoupled power flow algorithm based on a quantum state

that utilizes Hermitian and constant Jacobian matrices. Also,

a computationally improved Harrow-Hassidim-Lloyd (HHL)

algorithm is presented to deal with the fast decoupled power

flow by quantum phase estimation and reciprocal rotation. a DC power flow problem is solved using the HHL

algorithm, and it was demonstrated with a faster convergence

rate that the grid security can be enhanced using QC techniques.

However, the next generation quantum computers with large

quantum depths, qubit numbers, long coherence time, and lower

noises may be required to apply such approaches in large-scale

power systems, as suggested in 3.2 Optimal PMU placementOptimal phasor measurement unit (PMU) placement is a technique for reducing the number of phasor measurement units

(PMUs) necessary while ensuring that the complete power system is fully visible. The power system is identified as observable when the voltages of all buses in it are known. the

feasibility of QC is studied in solving optimal PMU placement optimization problems."

"Although QC provides solutions corresponding to complex

particle rotations, there are multiple sources through which

errors get added on to the results. For instance, there is a possibility that any two particles would not rotate to the same degree

when entangled. This is mainly due to the limited quality of

the control signals and the qubits sensitivity. Although, qubitsare kept in an isolated environment that restricts unexpected

movements, they tend to interact with the environment and

as a result, possibly lose the stored information along with

the entanglement. This process is termed as decoherence

and is being compensated using a quantum error correction

approach, which is also used to reduce quantum noise (that

get incorporated into the microwave signals). However, this

approach limits the number of qubits and the amount of

processing that can be done with them .

To put this study into perspective, since the information

pertaining to the state of the particle is expressed as a qubit, a

particle in a standby mode could be considered to be analogous

to an electrochemical battery, but it requires qubit operations

that comply with the postulates of QM, in order to absorb/extract energy (charge/discharge). Towards this, the quantum circuits and algorithms tailoring capability using Python allows battery researchers to perform P-QB simulations

including battery parameter predicting/forecasting

by making customized adjustments. Proper designing and

application of work extraction protocols, ways to maximize

the quantum entanglement, and quantum advantage will have

a positive impact towards extracting maximum benefit from

P-QB simulations. This is still an early stage for quantum algorithms to make drastic changes. Although multiple QC's have already been developed or are in the process, accessiblequbits restriction is a major limiting factor for both P-QB and C-QB simulations. This paper provides a review of the concepts linking particles' QM, and QC approaches together, for their application in the development of QBs. The approach that QB manufacturers would apply to extend its scalability to large scale applications still remains unknown. Finally, it is evident that the development and further advancements in QBs is closely linked with the developments in QC, hence the authors are of the impression that although this is an early stage for any QC driven innovations leading to formation of next generation energy storage technologies, the works discussed in this review indeed lay the foundation for future improvements."